

Task 14: Linux Server Hardening & Secure Configuration

Chapter 1: Introduction to Linux Server Hardening

Linux server hardening is the process of securing a Linux system by reducing its attack surface and configuring it according to security best practices.

This involves disabling unnecessary services, enforcing strict access controls, securing authentication mechanisms, and continuously monitoring system activity.

The goal of server hardening is to protect systems from unauthorized access, privilege escalation, data breaches, and common attack vectors such as brute-force attacks and misconfigurations.

Chapter 2: Review of Default System Settings

By default, Linux systems run multiple background services and have predefined user accounts and permissions. Reviewing default system settings helps administrators understand:

- Active users and groups
- Running services
- Open network ports

Unnecessary default configurations can introduce security risks if not reviewed and controlled. Therefore, identifying baseline system behaviour is the first step toward hardening.

Figure 2.1: *List of active services on the Linux system*

```
(ravnyx@hope)~$ systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
accounts-daemon.service            loaded active running Accounts Service
colord.service                      loaded active running Manage, Install and Generate Color Profiles
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
haveged.service                    loaded active running Entropy Daemon based on the HAVEGE algorithm
lightdm.service                    loaded active running Light Display Manager
ModemManager.service               loaded active running Modem Manager
NetworkManager.service             loaded active running Network Manager
polkit.service                     loaded active running Authorization Manager
rtkit-daemon.service               loaded active running RealtimeKit Scheduling Policy Service
systemd-journald.service            loaded active running Journal Service
systemd-logind.service              loaded active running User Login Management
systemd-udevd.service              loaded active running Rule-based Manager for Device Events and Files
udisks2.service                    loaded active running Disk Manager
upower.service                     loaded active running Daemon for power management
user@1000.service                  loaded active running User Manager for UID 1000
virtualbox-guest-utils.service      loaded active running Virtualbox guest utils

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
          ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
          SUB    → The low-level unit activation state, values depend on unit type.

18 loaded units listed.
```

Figure 2.2: *Open TCP and UDP ports using ss command*

```
(ravnyx@hope)~$ ss -tln
Netid      State      Recv-Q     Send-Q     Local Address:Port      Peer Address:Port
```

Chapter 3: User Account Management & Least Privilege

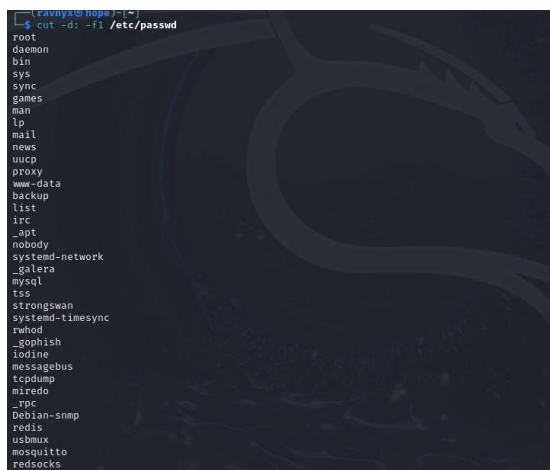
The **principle of least privilege** ensures that users are granted only the permissions necessary to perform their tasks.

Key practices include:

- Reviewing existing user accounts
- Removing unused or inactive accounts
- Limiting sudo privileges to trusted users only

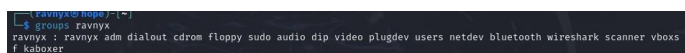
Restricting administrative access reduces the risk of accidental system damage and privilege abuse.

Figure 3.1: *List of user accounts from /etc/passwd*



```
[~] (ravnyx@hope) ~  
└─$ cat -d: -f: /etc/passwd  
root:  
daemon:  
bin:  
sys:  
sync:  
games:  
man:  
lp:  
mail:  
news:  
uuucp:  
proxy:  
www-data:  
backup:  
list:  
irc:  
_apt:  
nobody:  
systemd-network:  
_galera:  
mysql:  
tss:  
strongswan:  
systemd-timesync:  
rshod:  
_gophish:  
_odine:  
messagebus:  
tcpdump:  
niredo:  
_rpc:  
Debian-snm:  
redis:  
usbmux:  
mosquitto:  
redsocks:
```

Figure 3.2: *Group membership of current user*



```
[~] (ravnyx@hope) ~  
└─$ groups ravnyx  
ravnyx : ravnyx adm dialout cdrom floppy sudo audio dip video plugdev users netdev bluetooth wireshark scanner vboxs  
f kaboxer
```

Chapter 4: Sudo Access Review and Privilege Control

The sudo mechanism allows controlled administrative access without sharing the root password.

Reviewing sudo permissions ensures:

- Only authorized users can execute privileged commands
- All privileged actions are logged for accountability

Proper sudo configuration helps prevent unauthorized system changes and supports auditability.

Figure 4.1: *Verification of sudo privileges*

```
(ravnyx@hope)-[~]
$ sudo -l
[sudo] password for ravnyx:
Matching Defaults entries for ravnyx on hope:
env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin, use_pty

User ravnyx may run the following commands on hope:
(ALL : ALL) ALL
```

Chapter 5: SSH Hardening and Secure Remote Access

Secure Shell (SSH) is commonly used for remote server administration and is a frequent attack target.

SSH hardening includes:

- Disabling direct root login
- Enforcing key-based authentication
- Running SSH only when required

These measures reduce the risk of brute-force attacks and unauthorized remote access.

Figure 5.1: *SSH service status before hardening & SSH service enabled and running after configuration.*

```
(ravnyx@hope)-[~]
$ sudo systemctl status ssh
o ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:ssh_config(5)

(ravnyx@hope)-[~]
$ sudo systemctl start ssh

(ravnyx@hope)-[~]
$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-02-10 13:08:13 IST; 4s ago
     Invocation: f2e40916e084c088bcbab79d553b65a
       Docs: man:sshd(8)
            man:ssh_config(5)
   Process: 20509 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 20510 (sshd)
       Tasks: 1 (limit: 10375)
      Memory: 1.0M (peak: 2.1M)
         CPU: 33ms
    CGroup: /system.slice/ssh.service
            └─sshd "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 10 13:08:13 hope systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 10 13:08:13 hope sshd[20510]: Server listening on 0.0.0.0 port 22.
Feb 10 13:08:13 hope sshd[20510]: Server listening on :: port 22.
Feb 10 13:08:13 hope systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

(ravnyx@hope)-[~]
$
```

Figure 5.2: *SSH configuration showing PermitRootLogin before disabling.*

```
File Actions Edit View Help
CPU nano 8.4 /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file. See
# sshd_config(8) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 22
AddressFamily any
ListenAddress 0.0.0.0
ListenAddress ::

HostKey /etc/ssh/ssh_host_rsa_key
HostKey /etc/ssh/ssh_host_ecdsa_key
HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#CiphersList default
#KeyExchange default

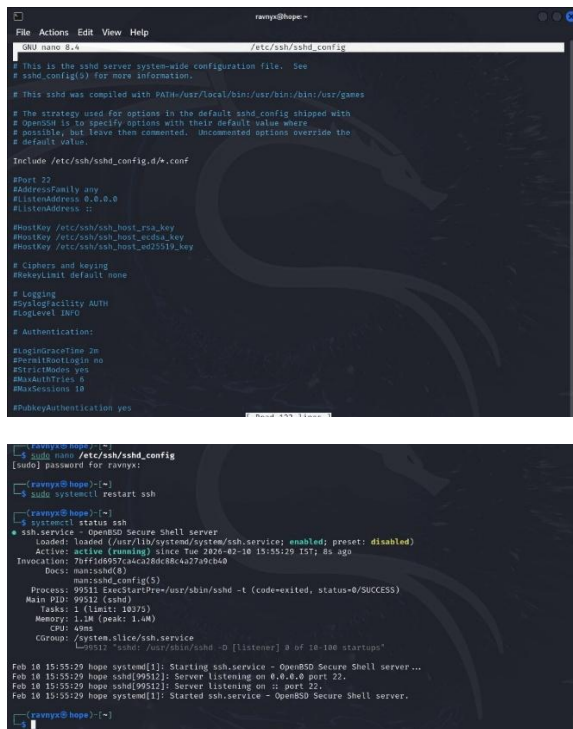
# Logging
SyslogFacility AUTH
LogLevel INFO

# Authentication:

LoginGraceTime 2m
PermitRootLogin prohibit-password
StrictModes yes
MaxAuthTries 6
MaxSessions 10

PubKeyAuthentication yes
```

Figure 5.3: *SSH service restarted after disabling root login.*



Chapter 6: System Updates and Patch Management

Regular system updates are critical for fixing known vulnerabilities and security flaws.

Updating packages ensures:

- Security patches are applied promptly
- Exploitable software bugs are removed
- System stability and compatibility are maintained

Keeping the system updated significantly reduces exposure to known attacks.

Figure 6.1: *Updating Linux package repository lists*

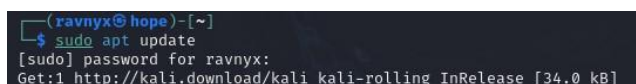
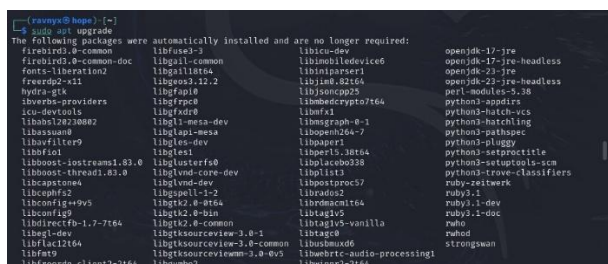


Figure 6.2: *Applying system package updates*



Chapter 7: Firewall Configuration and Network Traffic Control

Firewalls regulate incoming and outgoing network traffic based on predefined security rules.

Key firewall objectives:

- Allow only required ports and services
- Block unauthorized or suspicious connections
- Minimize external attack exposure

A properly configured firewall acts as the first line of defense against network-based attacks.

Figure 7.1: *Restricted inbound traffic configuration*

```
(ravnyx@hope)~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)

(ravnyx@hope)~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)

(ravnyx@hope)~$ sudo ufw status verbose
Status: inactive
```

Figure 7.2: *Firewall rules allowing only required services*

```
(ravnyx@hope)~$ sudo ufw enable
Firewall is active and enabled on system startup

(ravnyx@hope)~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
3306/tcp ALLOW IN Anywhere
Anywhere DENY IN 192.168.1.100
22/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
3306/tcp (v6) ALLOW IN Anywhere (v6)
```

Figure 7.3: *SSH allowed rule*

```
(ravnyx@hope)~$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)

(ravnyx@hope)~$
```

Chapter 8: Service Management and Attack Surface Reduction

Each running service increases the system's attack surface.

Hardening involves:

- Identifying running services
- Disabling unnecessary services
- Ensuring only essential services remain active

Reducing the number of active services limits potential exploitation points.

Figure 8.1: *Services review (no unnecessary services observed)*

```

(ravnyx@hope)~$ systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
accounts-daemon.service            loaded active running Accounts Service
cstore.service                     loaded active running Manager, Install and Generate Color Profiles
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
lightdm.service                    loaded active running LightDM daemon based on the XDMCP algorithm
lightdm.service                    loaded active running Light Display Manager
NetworkManager.service             loaded active running Network Manager
polkit.service                     loaded active running Authentication Manager
rftpd-daemon.service               loaded active running Rsync File Transfer Protocol Scheduler Policy Service
ssh.service                        loaded active running OpenSSH Secure Shell server
systemd-journald.service            loaded active running Journal Service
systemd-logind.service              loaded active running User Login Management
systemd-udevd.service               loaded active running udev Kernel Daemon for Device Events and Files
udisks2.service                    loaded active running Disk Manager
upower.service                     loaded active running Daemon for power management
user@1000.service                  loaded active running User Manager for UID 1000
virtualbox-guest-utils.service      loaded active running Virtualbox guest utils

legend: LOAD = Loaded, whether the unit definition has properly loaded.
         ACTIVE = The high-level unit activation state, i.e. operationalization of sub-
         SUB = The low-level unit activation state, values depend on unit type.

19 loaded units listed.

```

Chapter 9: File Permissions and Sensitive File Protection

Critical system files such as `/etc/passwd`, `/etc/shadow`, and `/etc/group` store authentication and user data.

Securing these files involves:

- Restricting read/write permissions
- Ensuring ownership by root
- Preventing unauthorized access

Proper file permissions protect sensitive system information from disclosure or manipulation.

Figure 9.1: Detailed metadata of `/etc/shadow` file

```

(ravnyx@hope)~$ stat /etc/shadow
File: /etc/shadow
Size: 1418      Blocks: 8      IO Block: 4096   regular file
Device: 8,1    Inode: 130971  Links: 1
Access: (0640/-rw-r-----)  Uid: (  0/   root)   Gid: ( 42/  shadow)
Access: 2026-02-10 12:17:35.716000000 +0530
Modify: 2025-04-28 14:05:40.827635284 +0530
Change: 2025-04-28 14:05:40.855621284 +0530
 Birth: 2025-04-28 14:05:40.823637285 +0530

```

Figure 9.2: Permissions of critical system files

```

(ravnyx@hope)~$ ls -l /etc/passwd /etc/shadow /etc/group
-rw-r--r-- 1 root root 1293 Apr 28 2025 /etc/group
-rw-r--r-- 1 root root 3197 Apr 28 2025 /etc/passwd
-rw-r----- 1 root shadow 1418 Apr 28 2025 /etc/shadow

```

Chapter 10: Log Monitoring and System Auditing

System logs provide visibility into authentication events, service activity, and system errors.

Log monitoring helps:

- Detect failed login attempts
- Track sudo and SSH usage
- Identify abnormal system behavior

Regular log review is essential for incident detection and forensic analysis.

Figure 10.1: System authentication and activity logs

- Patch management
- Service minimization
- Continuous monitoring

Applying these best practices significantly strengthens system resilience against common threats.

Chapter 12: Conclusion

Through systematic hardening, the Linux system was secured against common misconfigurations and attack vectors.

By enforcing least privilege, securing SSH access, controlling services, protecting sensitive files, and monitoring logs, the system achieves a stronger security posture suitable for real-world environments.